

WARNING!

USE A USB CABLE WITHOUT THE 5 VOLT PIN

REMOVE THE POWER PIN (PIN 1, 5 V) AND CHECK
WITH A MULTIMETER THAT 5 VOLTS DO NOT PASS.

OTHERWISE THE MISTER CAN BREAK!

OutRunners for two: simple guide

How to link two MiSTer FPGAs and try the test core (alpha).

1. First of all: the cable

You need a USB 3.0 cable with a **Type-A plug on both ends, short** (tested at 30 cm) and of good quality, to be connected to the **USER port** of both MiSTers (the blue USB port on the I/O board, which is not a real USB port). Joining the 5 V of two MiSTers can damage them: that is why the 5 V wire (**pin 1, called VBUS**) **must not be connected**. The other wires stay as they are.

How to remove the 5 V (any one of these is enough)

- 1 **Cut the red wire (VBUS)** inside the cable, if you can open it.
- 2 **Break or remove pin 1** inside **one** of the two USB-A plugs.
- 3 Use a **small pass-through adapter** that has no pin 1.

Multimeter test: mandatory

The multimeter must read 0 V. If you read about 5 V the cable still passes power: DO NOT use it.

- 1 Plug **only one end** of the cable into a **USB charger** (or power bank). The other end stays free.
- 2 Multimeter on **DC volts**, 20 V range.
- 3 The free plug has **4 long contacts** (the old USB 2.0 ones) and, further back, 5 short ones. Measure **every pair of the 4 long contacts** (6 pairs): they must always read **0 V**. Do not touch two neighbouring contacts with the same probe: short circuit.
- 4 If you read about 5 V, remove the pin and repeat the test. When finished, **unplug the cable from the charger** before connecting it to the MiSTers.

2. What you need

What	Notes
2 MiSTer FPGAs	Both with the USER port. With a SuperStation One and its SuperDock, set the SNAC Bypass switch to ON (as in our tests).
1 USB 3.0 A-A cable	Without the 5 V pin (see page 1), short, good quality.
OutRunners ROMs	Not included. Use the same set that already works with the original MRA.
The files of this guide	outrunners.rbf (the core) and two .mra files (World and US).
No SNAC attached	Nothing else must be plugged into the USER ports.

3. Installing on the two MiSTers

Do the same on **both** MiSTers (same files on each).

- 1 Power off the MiSTer and take out the SD card (or connect to the MiSTer over the network).
- 2 Copy **outrunners.rbf** into the **_Arcade/cores/** folder.
- 3 Copy the two **.mra** files into the **_Arcade/** folder:
 - OutRunners (World) Link.mra
 - OutRunners (US) Link.mra
- 4 Check that the ROMs are in place (usually **games/mame/**).
- 5 **Important:** in the **_Arcade/cores/** folder there must be **only one file starting with «outrunners»**. Move any old copies elsewhere, otherwise the MiSTer may start the wrong one.

To be sure the file is the right one, the checksum (md5) of outrunners.rbf must be **5da6d67d9c9b9594b859f3c413dfc1b5** (4592880 bytes). The right core shows the entries «Comm RAM Clear» and «Comm Hi Byte» in the OSD.

4. Connecting the cable

- 1 **Power off both MiSTers** before connecting the cable.
- 2 Connect the cable to the **USER port (blue USB)** of each MiSTer. Do not use the normal USB ports for controllers.
- 3 Power the two MiSTers back on.

5. OSD settings (on both machines)

From the MiSTer menu start **OutRunners (World) Link** or **OutRunners (US) Link** (use the same version on both). Open the OSD (F12 key or the button you mapped) and set:

OSD entry	Value	Why
Cabinet Link	Network	Turns on the link between the two MiSTers.
Link Baud	250k	Link speed. Same on both.
Link Test	Off	It is a cable test (see later): off while playing.
Comm RAM Clear	Once	Already the default: leave it.
Comm Hi Byte	FF	Already the default: leave it.
Link Role / Cabinet ID	leave as they are	The core ignores them: role and number are decided by the game (step 6).

If you do not see «Comm RAM Clear» and «Comm Hi Byte», the MiSTer is using an old core: check step 3.

6. Settings inside the game (Service / Test menu)

The game stores these settings in its own memory (EEPROM): do them **once** on each MiSTer. If you rename the .mra file, the game settings may start again from zero and must be redone.

- 1 Enter the game's **Test menu** the way you usually do (the OSD «Service Mode» option or the Test button on your controller).
- 2 In the list choose **NETWORK ASSIGNMENTS**. As the on-screen text says: «**SELECT BY SERVICE BUTTON AND PUSH TEST BUTTON**» (the Service button picks the row, the Test button confirms/changes).
- 3 Set **COMMUNICATION = NETWORK**. Two more rows appear.
- 4 Set **PRIVILEGE MODE** and **CABINET ID#** as in the table below.
- 5 Choose **EXIT** and leave the Test menu.

Game entry	MiSTer 1 (Master)	MiSTer 2 (Slave)
COMMUNICATION	NETWORK	NETWORK
PRIVILEGE MODE	MASTER	SLAVE
CABINET ID#	1	2

One MiSTer must be Master and the other Slave. Do not use the ISDN/Relay mode: it is not supported.

7. Starting the link

- 1 Check that both games are set to **Network** (step 6) and both MiSTers to **Cabinet Link = Network** (step 5).
- 2 **Reset from the OSD on both machines**, a few seconds apart.
- 3 The **NETWORK CHECK** screen appears with the network checks.

If all goes well, each MiSTer shows this screen:

NETWORK CHECK
COMMUNICATION SUCCESS.
THIS MACHINE ID IS **1** (or **2**)
THIS MACHINE IS **MASTER CONTROLLER** (or **SLAVE**).

Then you can start playing. In our test the link stayed up for more than 30 minutes.

8. The LED that tells what is going on

With **Link Test = Off**, the disk (HDD) LED on the MiSTer I/O board counts how far the link has got: it flashes as many times as the step number, then pauses, then starts again. (The SuperStation may have no visible LED: in that case watch the screen.)

Flashes	Meaning
0 (always off)	The game has not turned the network board on (check step 6).
1	Board turned on.
2	The game wrote its signature and we answered.
3	The game read the answer.
4	The game wrote whether it is Master or Slave.
5	Valid role: messages start going to the other MiSTer.
6	A message arrived from the other MiSTer.
7	Link up.
Fast flicker	The link was up and stopped for more than 3 seconds.

9. Cable test (Link Test)

If the link does not start, first test only the cable, without the game:

- 1 On **both** MiSTers: **Cabinet Link = Network, Link Test = On**, Link Baud 250k.
- 2 The MiSTer LED must stay **steadily on**: it means messages are arriving from the other one.
- 3 Unplug the cable: the LED must go off after about 1 second. Plug it back: it comes on again.
- 4 When finished set **Link Test = Off** again.

If the LED does not come on with Link Test: try Link Baud 115k or 57.6k, **the same on both**, check you used the USER port (not a normal USB one) and try another, shorter cable.

10. If something goes wrong

What you see	What to do
NETWORK ERROR / CONNECTION ERROR right away	Check that the OSD shows «Comm RAM Clear» and «Comm Hi Byte» (right core), that there is only one outrunners file in the cores folder, and that both MiSTers are in Network (OSD and game).
LED always off	The game has not turned the network on: redo step 6 and leave the menu with EXIT.
LED at 5 flashes but not 6 or 7	The other MiSTer does not answer: check the cable, that one is Master and the other Slave, that Link Baud is the same, and Reset a few seconds apart.
COMMUNICATION TIME OUT	One of the two waited too long: Reset both a few seconds apart.
LED fast flicker	The link dropped: check the cable and Reset.
After changing the .mra file the game went back to Stand-alone	The game settings may be tied to the file name: redo step 6.

Limits of this test version

- Works only with **two** MiSTers (one Master and one Slave).
- The link uses our own system: **it is not compatible with original cabinets**.
- The link and the «COMMUNICATION SUCCESS» screen work; **it is not yet verified in a race** that the other player's car moves correctly. Tell us what you see.

How to report a result

Write: which game (World or US), which MiSTer is Master and which is Slave, what each screen shows after Reset, how many times the LED flashes, and whether in a race you see the other car (and whether it moves smoothly or in jerks).

Base core: Meathax (s32multi). Link code: github.com/bonna79/s32multi-mycore (code version: commit 4fc543f). The cable and pins are the same as in Kuba-J's «PSX link cable» (alpha) core, with which the cable was tested. Test version: it may change.